

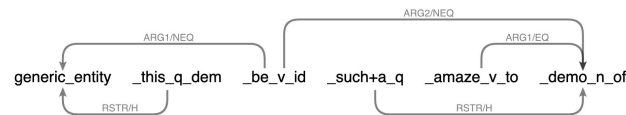
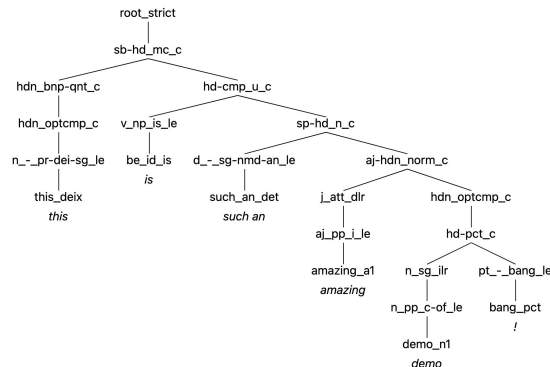
DELPH-IN Summit 2026

Mandarin Resource Grammar

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TOP INDEX	$h0$ $e2$				
RELS	$\left\langle \begin{array}{l} \text{generic_entity}(0:4) \\ \text{LBL } h4 \\ \text{ARG0 } x3 \end{array} \right\rangle$	$\left\langle \begin{array}{l} \text{_this_q_dem}(0:4) \\ \text{LBL } h5 \\ \text{ARG0 } x3 \\ \text{RSTR } h6 \\ \text{BODY } h7 \end{array} \right\rangle$	$\left\langle \begin{array}{l} \text{_be_v_id}(5:7) \\ \text{LBL } h1 \\ \text{ARG0 } e2 \\ \text{ARG1 } x3 \\ \text{ARG2 } x8 \end{array} \right\rangle$	$\left\langle \begin{array}{l} \text{_such+a_q}(8:15) \\ \text{LBL } h9 \\ \text{ARG0 } x8 \\ \text{RSTR } h10 \\ \text{BODY } h11 \end{array} \right\rangle$	$\left\langle \begin{array}{l} \text{_amaze_v_to}(16:23) \\ \text{LBL } h12 \\ \text{ARG0 } e13 \\ \text{ARG1 } x8 \\ \text{ARG2 } i14 \end{array} \right\rangle$
HCONS	$\left\langle \begin{array}{l} \text{qeq } h0 \\ \text{HARG } h1 \\ \text{LARG } h1 \end{array} \right\rangle$	$\left\langle \begin{array}{l} \text{qeq } h6 \\ \text{HARG } h4 \\ \text{LARG } h4 \end{array} \right\rangle$	$\left\langle \begin{array}{l} \text{qeq } h10 \\ \text{HARG } h12 \\ \text{LARG } h12 \end{array} \right\rangle$		

Summary

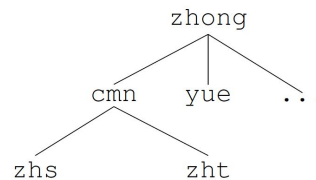
- Brief History and Renaming Motivation
- Quick rundown of the state-of-affairs
- Regression Testing (incl. new test suites)
- Plans for the future

Have you hear of Post-DELPH-IN Blues?

It turns out that:
Grammars make linguists happy...
And LLMs don't...

Brief History and Renaming Motivation

- Zhong [Fan, Song, Bond, 2015]
 - Mandarin Chinese grammar (MCG) [Zhang, Wang, Chen, 2011]
 - ManGO [Yang, 2007]



- Goal: a Grammar Sharing/ Metagrammar Engineering / CLIMB implementation for Chinese languages
- Reality: imposed varied configuration/symlinking that quickly broke down once only Mandarin Chinese was being developed
- New Cantonese Grammar was also born outside Zhong

Zhong is dead, long live the MRG

- The name **Mandarin (Chinese) Resource Grammar (MRG)**
 - Auspicious, Aspirational, Simple & Findable
- It primarily aims to be a resource for linguists and L2 learners
- Interesting opportunities to grow alongside the Cantonese grammar
- Focus more on syntactic coverage, and less on lexicon coverage

Lexicon

MRG inherits a messy lexicon:

- **lexicon.tdl** [~17k entries] Large is automatically acquired (FZZ)
- **lex-numbers.tdl** [~20k entries] Missing a treatment of numbers
- **lex-classifiers.tdl** [454 entries]
- **lex-symbols.tdl** [61 entries]

New world:

- **lexicon-core.tdl** [~600 entries] Hand-checked, moved from lexicon.tdl
- **lex-base.tdl** [~250 entries] HSK-graded (linked); with romanization, English definitions; (soon to be linked to wordnet)

Lexicon

```
=====
; HSK1_WID1: 爱 [ài]
;=====
爱_v_1 := v_np-spr_le &
=====
[ài] to love; to like; to be fond of;
=====
[ STEM < "爱" >,
  SYNSEM.LOCAL.CAT.HEAD.CHAR [ FCHAR "爱", LENGTH one ],
  SYNSEM [ LKEYS.KEYREL.PRED "_爱_v_1_rel" ],
  TRAITS native_token_list ].

爱_v_2 := v_vp_sequi-spr_le &
=====
[ài] to be inclined (to do sth);
=====
[ STEM < "爱" >,
  SYNSEM.LOCAL.CAT.HEAD.CHAR [ FCHAR "爱", LENGTH one ],
  SYNSEM.LKEYS.KEYREL.PRED "_爱_v_2_rel",
  TRAITS native_token_list ].

爱_n_1 := n_-le &
=====
[ài] love; affection;
=====
[ STEM < "爱" >,
  SYNSEM.LOCAL.CAT.HEAD.CHAR [ FCHAR "爱", LENGTH one ],
  SYNSEM.LKEYS.KEYREL.PRED "_爱_n_1_rel",
  TRAITS native_token_list ].

爱_n_2 := n_-pn_le &
=====
[ài] common proper name;
=====
[ STEM < "爱" >,
  SYNSEM.LOCAL.CAT.HEAD.CHAR [ FCHAR "爱", LENGTH one ],
  SYNSEM.LKEYS.KEYREL.CARG "爱",
  TRAITS native_token_list ].
```

- The New HSK vocabulary is surface-form based
- If/when I have information about senses, I would like to restrict things a bit better

The Future Lexicon

- My goal is focus on and go over the HSK lexicon (gradually)
- I am very tempted to create some form of database/external tool to help me manage and link it to wordnet

Pearls of wisdom are welcome!

- I'm also interested in evaluating how LLMs perform in lexicon acquisition (e.g., as a sequence labelling task)

Proficiency Level (new)	No. of Surface Forms
HSK 1 (A1)	500
HSK 2 (A2)	772
HSK 3 (A2+)	973
HSK 4 (B1)	1000
HSK 5 (B2)	1071
HSK 6 (C1)	1140
HSK 7-8-9 (C2)	5636
Total	11092

Co-development of the Chinese Open Wordnet

- “Great minds think alike”

(Although I’m pretty sure that this is how some doomsday cults start)

- I have not chosen the flavour of my poison:
 - Main goal: be a useful resource for linguists & L2 learners
 - I’m hedging my bets: see what happens with the ERG
 - I like the idea of coarse senses... not sure how yet
- I’m happy to consider a “small world” restricted up to a certain level
- I also want to make wordnets better resources for L2 learners

Grammar Modes

- mrg-gram, mal-gram
- char-gram & robust-gram (soon)
 - the default will be ‘strict’;
 - “robust” can include a few things (e.g., lexicon, bridging rules, etc.)
 - Probably will remove “mrg-utterance-gram.tdl” until needed

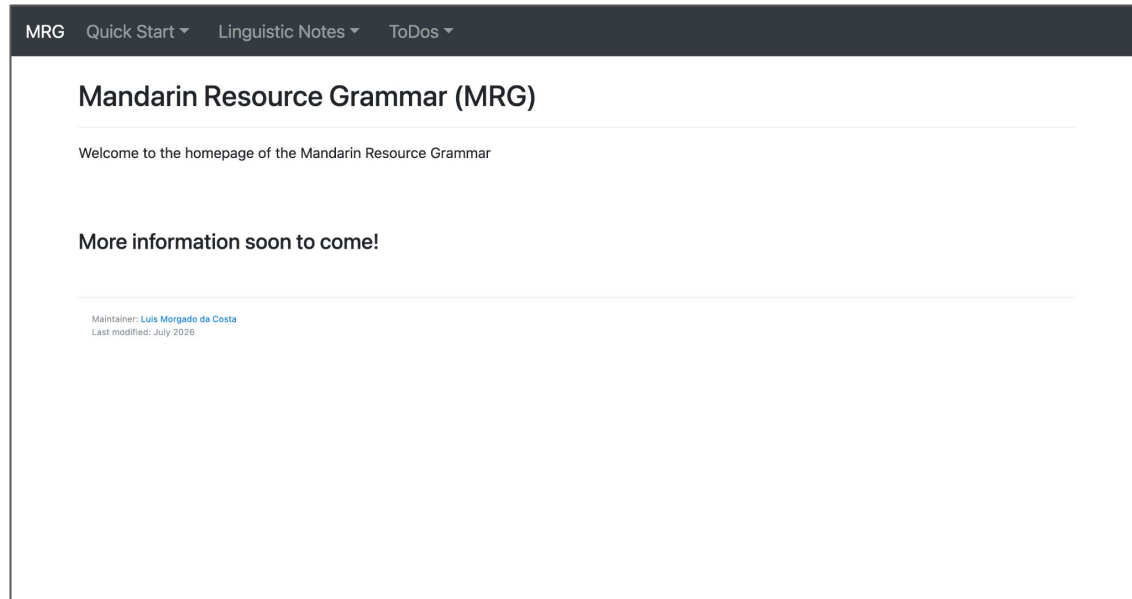
ACE and LKB-FOS compatible grammars

No new syntactic work, mostly lexicon, docstrings, spring-cleaning

Embedded Homepage

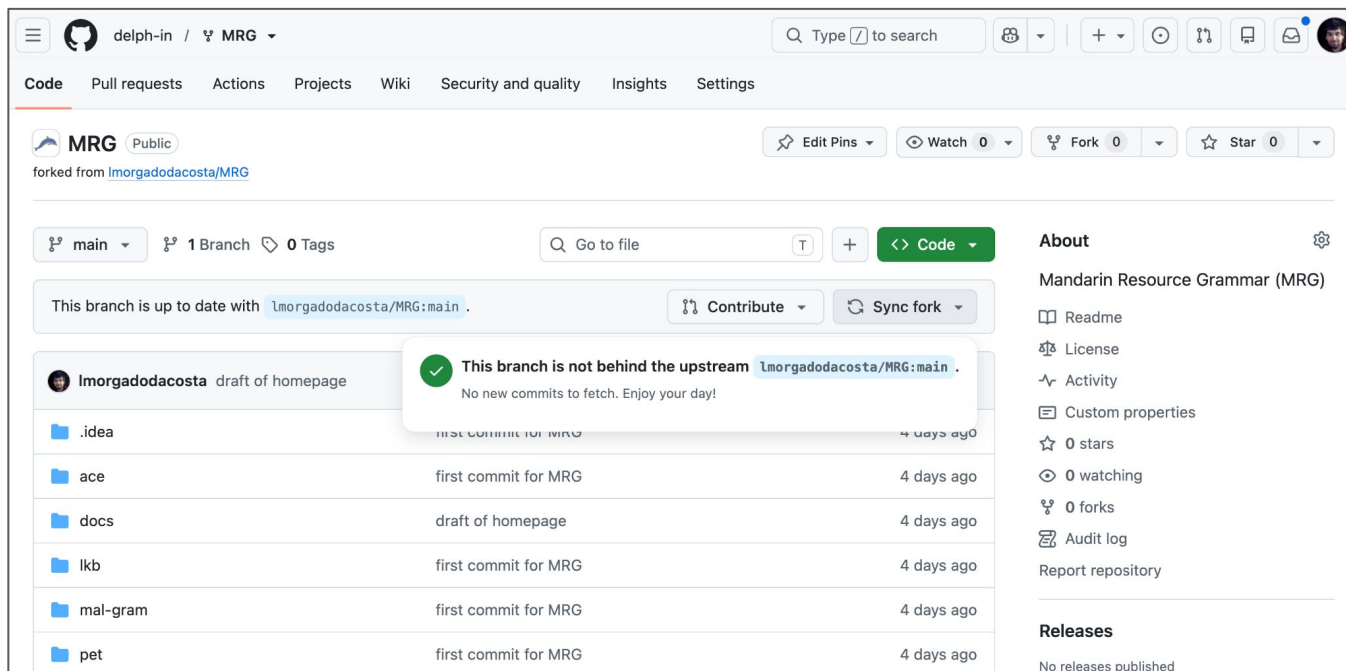
Documentation embedded: <https://lmorgadodacosta.github.io/MRG/>

It is also the grammar's homepage



Forked into DELPH-IN Github

Can be updated from its origin with a click of a button



Regression Testing (Big Picture)

- I dusted off some test suites and created a couple new ones
- Most of them are in the education domain
- No big surprised with results (almost A2 viable)

HSKStandardCourse1	92.0
HSKSandardCourse2	78.3
HSKSandardCourse3	60.1
HSKSandardCourse4A	32.8
HSKSandardCourse4B	27.2

Regression	No. of Items	% Parsed
MRS	107	93.46%
CMN-Edu	798	92.73%
TUFS	1523	62.77%
Sinica	1946	47.53%
Tatoeba	47216	-
ChineseGrammarWikiA1	607	81.05%
HSKStandardCourse	1910	51.52%
LE (Phenomenon)	134	57.46%
A-NOT-A (Phenomenon)	107	50.47%
Classifiers (Phenomenon)	15	93.33%

Basics

Grammar Point (English)	Pattern	Examples
Basic sentence order	Subj. + Verb (+ Obj.)	我 爱 你 。
Connecting nouns with "shi"	A + 是 + B	我 是 学 生 。
Expressing "excessively" with "tai"	太 + Adj. + 了	太 好 了 ！
Simple "noun + adjective" sentences	Noun + 很 + Adj.	我 很 好 。

Question Forms

Grammar Point (English)	Pattern	Examples
Affirmative-negative question	Verb + 不 + Verb / Adj. + 不 + Adj.	去 不 去 ？ 好 不 好 ？
Placement of question words	什么 / 什么时候 / 谁 / 哪儿 / 为什么 / 怎么	你 是 谁 ？ 你 在 哪 儿 ？
Tag questions with "bu"	……是不是 / 对不对 / 好不好 ？	你 们 明 天 来 ， 好 不 好 ？
Tag questions with "ma"	……是吗 / 对吗 / 好吗 ？	这 样 做 ， 对 吗 ？
Yes-no questions with "ma"	……吗 ？	他 是 老 板 吗 ？

<https://resources.allsetlearning.com/chinese/grammar>

- Sufficiently open library (CC-BY-SA-NC)
- Up to B2
- Creates some problems in how to share some of the test suites
(not a new problem)

ChineseGrammarWikiA1

Affirmative-negative question	89.7
Age with "sui"	10.0
Comparing "er" and "liang"	69.6
Connecting nouns with "shi"	53.3
Expressing "and" with "he"	87.5
Expressing "be going to" with "yao"	50.0
Expressing "excessively" with "tai"	80.0
Expressing a learned skill with "hui"	100.0
Expressing ability or possibility with "neng"	66.7
Expressing close possession without "de"	90.0
Expressing existence in a place with "zai"	90.0
Expressing existence with "you"	90.0
Expressing permission with "keyi"	60.0
Expressing possession with "de"	100.0
Expressing possession with "you"	90.0
How to do something with "zenme"	54.5
Measure word "ge"	50.0
Negation of "you" with "mei"	88.2
Negation of past actions with "meiyou"	91.7

Negative commands with "buyao"	100.0
Offering choices with "haishi"	90.0
Placement of question words	83.1
Questions with "ne"	78.9
Simple "noun + adjective" sentences	84.6
Standard negation with "bu"	75.0
Structure of dates	90.0
Structure of days of the week	60.0
Structure of numbers	98.3
Structure of times (basic)	100.0
Suggestions with "ba"	70.0
Tag questions with "bu"	90.0
Tag questions with "ma"	70.0
The "all" adverb "dou"	80.0
The "also" adverb "ye"	80.0
Using the verb "jiao"	54.5
Using the verb "qu"	100.0
Using the verb "xing"	66.7
Wanting to do something with "yao"	88.9
Yes-no questions with "ma"	95.5

Plans for Upcoming Year

- Make MRG go “vroom vroom”
 - More parses = more better! (Singapore style)
 - Go through vocab sequentially (HSK1 & HSK2)
 - Try not to miss basic phenomena

(Joanna and I have agreed to work on comparative constructions)
- I will write an NWO “eXtra Small” (XS) grant (~50k) to work on “visual parsing for Grammatical Error Detection” – I’ll include English and Mandarin Chinese
- I would love to revive the “Grammar Engineering Meetings”

Thank you!



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